



Poincaré

The pool that knows when the market is really moving and leans against it.

Most AMMs quote the same way in a crash as they do on a quiet Sunday. Poincaré watches its own price, runs a **change-point detector** with a mathematical optimality proof behind it, and only defends its liquidity providers when a genuine trend is underway. No oracle, no keeper, and nothing to bribe.

◆ Manipulation costs more than it pays, by construction ◆ 118 tests green ◆ Proven on six months of real market data

THE PROBLEM

Liquidity providers lose money precisely when markets trend

There is an invisible tax on every pool, and it has a name: **Loss-Versus-Rebalancing**. Whenever the price moves, arbitrageurs trade against the pool's stale quote and keep the difference. The Milionis–Moallemi–Roughgarden–Zhang result tells us exactly what drives it:

$$\text{LVR rate} \approx \frac{1}{2} \cdot \sigma^2 \cdot \text{marginal liquidity}$$

Notice what is missing from that equation: **the fee**. Every fee-tweaking hook is pulling a lever that simply is not in the formula. And notice what drives it: sustained, one-directional moves. **Chop barely hurts LPs. Trends drain them.**

A normal pool during a trend

The curve is frozen, so it offers the same terms to the arbitrageur draining it as to the trader helping it. LPs pay the full tax, block after block, until the trend ends.

What would actually help

A pool that can tell a real trend from noise, and defends itself only while one is happening.

That judgement call is the hard part, and it is the whole project

"Has a real trend started?" was solved in 1954. Nobody brought it on-chain.

Quickest Change Detection

Statisticians have spent seventy years on exactly this question: how do you detect a regime change as fast as possible without crying wolf? Their workhorse answer is the **CUSUM statistic**. It quietly accumulates evidence every step and fires the moment that evidence crosses a threshold.

$$S_t = \max(0, S_{t-1} + \ell_t - k) \rightarrow \text{alarm when } S_t \geq h$$

And it comes with a proof: under Lorden's criterion, CUSUM minimises the worst-case detection delay for any tolerated false-alarm rate. This is not a heuristic that felt right. It is the best possible answer to the speed-versus-false-alarms question.

Why an AMM should care

The firing moment depends on the data, not the clock. A strong trend crosses the threshold in a few blocks, while noise never gets there at all. So there is **no fixed countdown** for an attacker to study and game.

Nobody else does this

We scanned the entire 562-hook UHI directory and found **zero** uses of CUSUM or any change-point method. Everything out there triggers on moving averages and fixed thresholds, which is exactly the kind of signal an attacker can predict.

To our knowledge, Poincaré is the **first AMM whose behaviour is governed by a provably optimal sequential change detector**.

A brain that decides, a curve that acts, a fee that breathes

The brain

Two-sided CUSUM detector

It runs on the pool's own log-returns, sampled once per block from pre-swap reserves so flash tricks never reach it. A directional-efficiency gate confirms the move is a march rather than a thrash, and clipped increments keep fat-tailed crypto returns from fooling it.

The actuator

A directional spread that LPs keep

When a trend is confirmed, flow pushing **with** the trend pays a small, capped spread, and that value goes straight to the LPs it would otherwise have been taken from. Flow that stabilises the pool trades at the full base price, because punishing your friends is bad business.

The revenue

A fee generated from volatility

In calm markets LPs still earn: the base fee is computed each block from the pool's own realized volatility, so it tightens when things are quiet and widens when they are not. Never a constant someone picked in a meeting.

Once per block:

pool price → log-return → CUSUM + gate → bounded, rate-limited κ → spread on the toxic side only

The signal is the pool's own price and nothing else. No oracle to corrupt, no keeper to bribe, no AVS to trust, and every parameter is injected and calibrated rather than hard-coded.

It ignores the noise, then commits when the evidence is real

Calm

Evidence hovers near zero, so the curve stays symmetric and deep. Everyone gets the best price the pool can offer.

A trend begins

The statistic starts climbing, but the detector holds its fire because early moves look exactly like noise. Crying wolf here would tax honest traders for nothing.

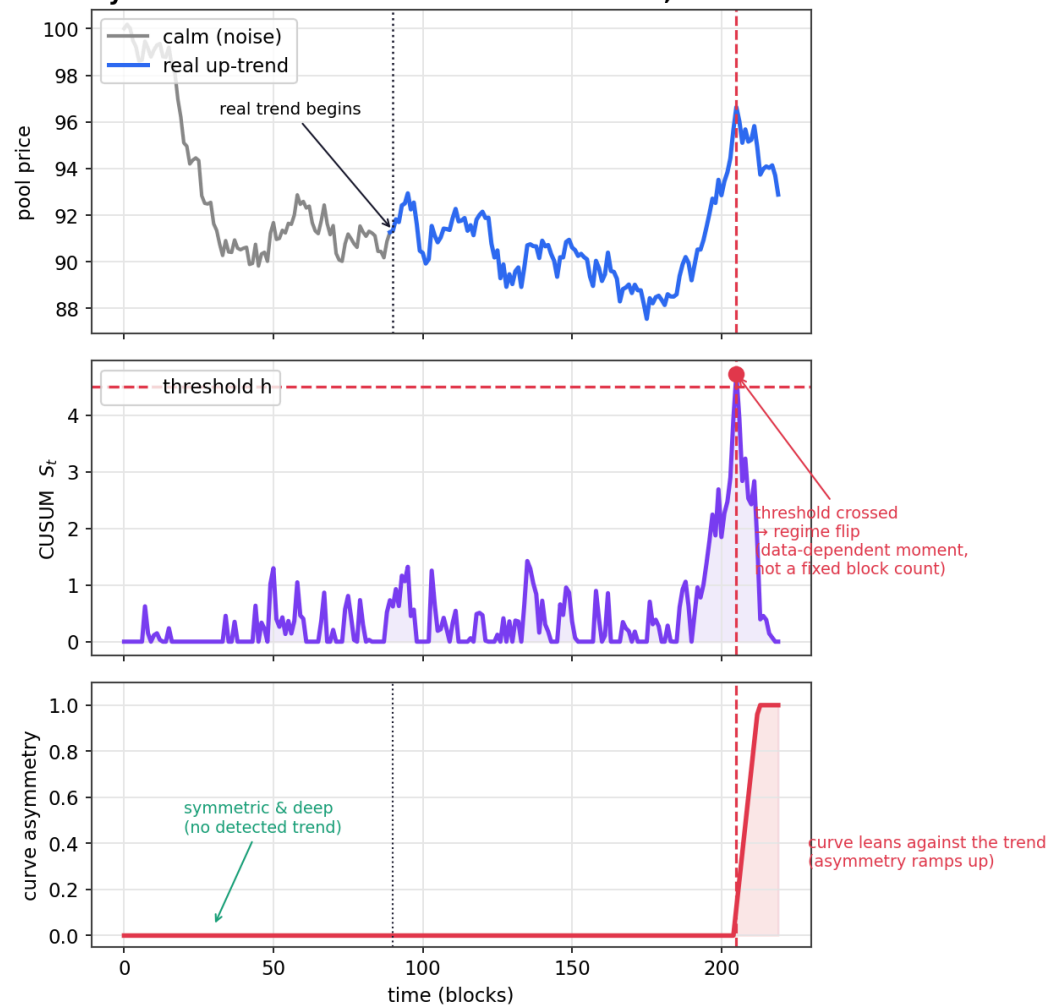
Detection

Evidence crosses the threshold at a moment set by the data, and only then does the curve lean, capped and rate-limited so it can never snap.

The trend fades

Returns shrink, the gate closes, and the lean ramps back down with hysteresis so it never flaps.

Fig 2 — Lifecycle on a real trend: evidence accumulates, then the curve reacts



Faking a trend means genuinely moving the market, at your own expense

1 · No countdown to study

The firing time depends on the data, and the detector samples once per block from settled reserves, so a flash loan that unwinds inside a block is simply invisible to it.

2 · Nothing to win

The spread only ever makes trades worse, never better. The "soft" side trades at the plain base price, so an attacker who fakes a trend finds zero edge waiting on the other side.

3 · Arbitrageurs collect the toll

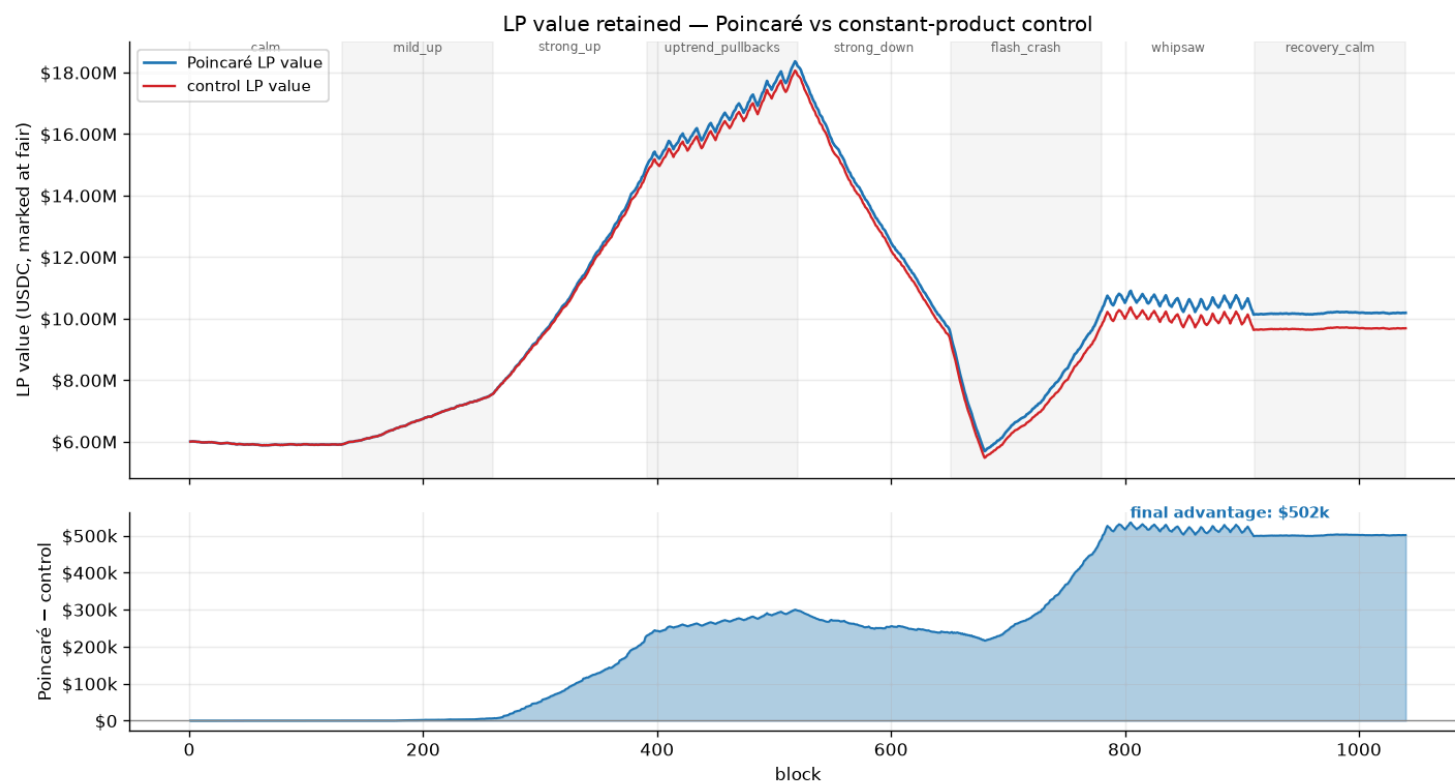
Pushing the price off fair value opens a gap that arbitrageurs harvest the moment the attacker stops. The chart shows their cost climbing the entire time, and it buys them nothing.

All three layers are simulated end-to-end against the real PoolManager in our test suite: fake-trend round trips lose money, flash spikes never reach the detector.

Fig 3 — A fake trend isn't fake: to move CUSUM the attacker must really move the market

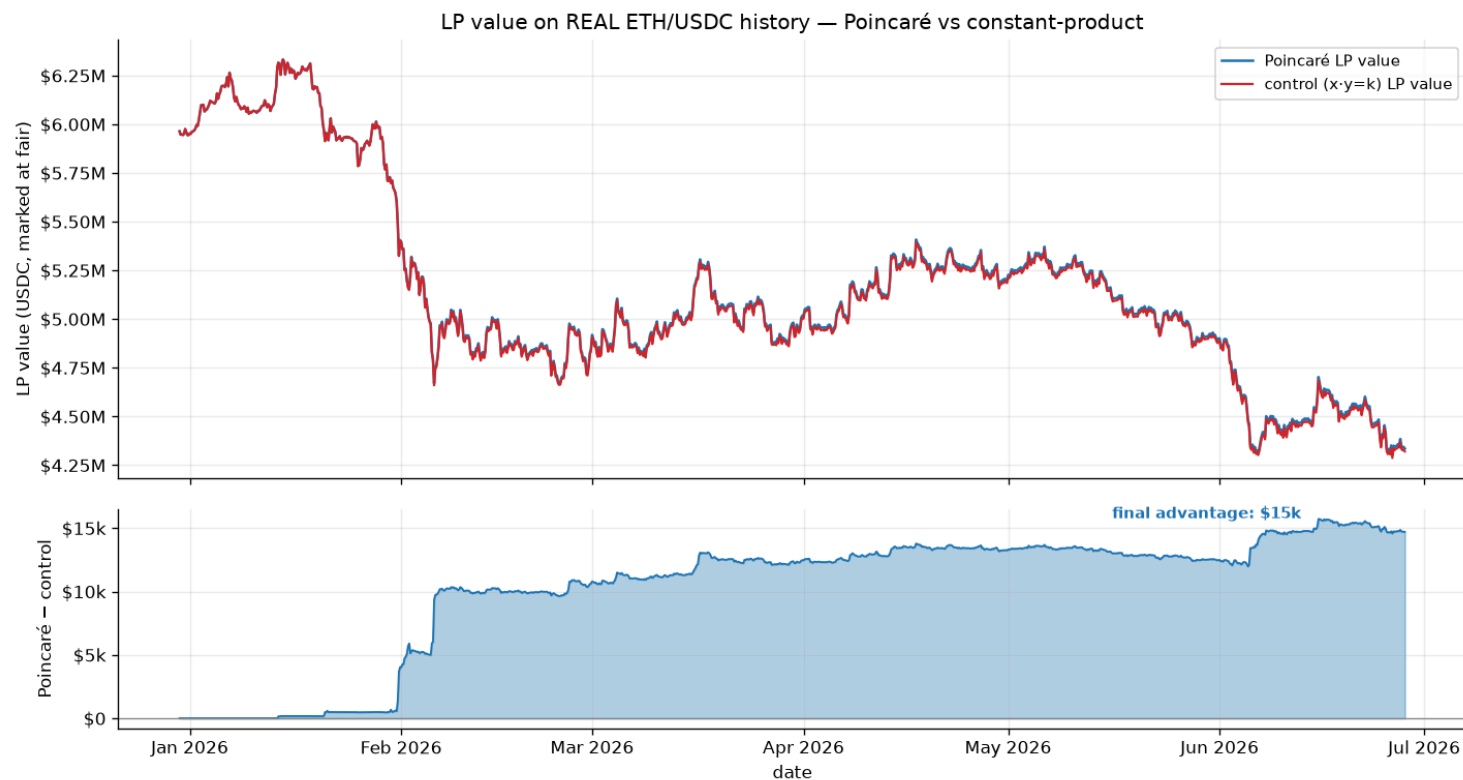


Same pool, same trades, one difference: the detector. LPs kept \$502k more.



Two identical pools on the real Uniswap v4 PoolManager, fed the same fair-price path and the same 3,842 swaps across eight regimes. Look at where the blue gap grows: it stays flat through calm, because the detector correctly does nothing, then it widens through every trend and jumps hardest in the flash crash, where LVR was cut by 83%. Protection shows up exactly where LPs bleed the most.

Six months of real ETH/USDC, through a 47% bear market. It never hurt once.



The same head-to-head, now driven by real Binance ETH/USDC closes from December 2025 to June 2026, a genuine multi-leg bear market. The advantage stays flat through choppy January, then steps up at the February crash and again at the June leg down, because those are the two real sustained trends the detector caught. LVR came in 8.8% lower than the control, and the Poincaré pool never once trailed it. That floor held through every single block, and we assert it in the test suite.

Measured, reproducible, and honest about where each number comes from

-14.3%

LVR vs constant-product
synthetic regime back-test

-29.6%

with the adaptive detector
thresholds in units of live σ , zero manual tuning

~2×

less tax on honest flow
vs a symmetric vol-fee with the same average spread

~96k

gas per sampled block
paid once per block, not per swap

Why it beats a volatility fee

A symmetric vol-fee defends LPs by taxing everyone, friend and foe alike. Poincaré spends the same friction budget only on the toxic side and only while a trend is confirmed, so it wins on LVR *and* charges honest traders about half as much. That is the frontier that matters: protection per unit of collateral damage.

The engineering behind the claims

- ✓ 118 Foundry tests, zero failures
- ✓ Two invariant suites, 128k random operations each, solvency exact to the wei
- ✓ Adversarial sims: fake trends, flash loans, volatility inflation
- ✓ Quotes match on-chain execution exactly, via the Lens

Running on a public testnet, carrying six months of real market history

179

real on-chain swaps

daily ETH/USDC closes replayed through the live pool, block by block

\$45,676

LVR captured for LPs

on \$5.7M of replayed volume

16

with-trend leans

at full strength during the Feb crash and the June leg down, and zero across three months of chop

And it's a product, not just a contract

- ▶ Every sampled block, the hook publishes its full state as an on-chain event: evidence, volatility, lean, regime and fee. The app charts the real statistics, so what you see is what the contract did.
- ▶ Quotes come from an on-chain lens that runs the exact pricing code the pool runs, so what you are quoted is what you get, to the wei.
- ▶ In our live demo the detector held quiet through twelve buys, fired on the thirteenth once the evidence was in, rode the trend at full lean, then unwound cleanly and re-fired on the reversal.
- ▶ The whole lifecycle from slide five, observable on a public chain by anyone.

DIFFERENTIATION

Everyone else reacts to a heuristic. Poincaré can prove its trigger is optimal.

AXIS	EXISTING HOOKS	POINCARÉ
Lever	fee tweaks on a static curve	trend-gated directional spread, arb-safe by construction
Trigger	fixed windows, thresholds, oracles	CUSUM stopping time, data-dependent and unpredictable
Optimality	felt right in a notebook	Lorden minimax-optimal detection
Manipulation	hopes the signal is hard to fake	zero prize on the soft side, plus arbitrage collects the attacker's toll
Calm-market LP income	a flat fee someone chose	a fee computed from realized volatility, every block
External dependencies	oracles, AVSs, keepers	none, and that is a security property

Routing takes care of itself: calm and stabilising flow gets the full base price, so best-execution routers send it here. Confirmed-toxic flow gets quoted worse on purpose, so it goes elsewhere. The pool sheds exactly the flow that costs LPs money and keeps the flow that doesn't.

What it does, and what we refuse to pretend

Where it shines

Liquid, free-floating, volatile pairs like ETH/USDC, because that is where directional LVR actually lives, the price series is clean, and moving the market costs real money. Pegged pairs degrade gracefully: the curve just stays symmetric, and a depeg is simply a trend it leans into.

It removes the avoidable arbitrage tax, not market risk itself. LPs still hold the assets, and we say so on the landing page rather than in a footnote.

We found our own exploit, and shipped around it

The obvious version of this idea, bending the curve's depth per direction, turns out to be drainable. We proved it with a concrete counterexample: a bot buys the deep branch, sells back on the shallow one, and walks off with pool value. So we shipped the spread construction, which is safe because it can only ever make a round trip worse, and we keep the depth lever on the roadmap behind the safety analysis it deserves.

Same discipline on the adaptive detector: it is built, simulated against volatility attacks, and it backtests at twice the LVR cut, but it will not guard real value until its worst-case bound is derived.

ROADMAP

From live MVP to production

Now

Calibrate per pair on real return series (the harness is already data-ready), and derive the adaptive detector's manipulation bound so its 29.6% cut can go live safely.

Next

Router and aggregator integration through the Lens, multi-pool coverage, and the depth lever if and only if its safety sizing proves out.

Then

External audit, then mainnet on the volatile majors, where LVR is measured in billions per year across DeFi.

The thesis in one line

The only way to fool the detector is to genuinely move the market, which is exactly what it exists to react to. So the protection is bounded by mathematics rather than wished into place.



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